

Parameters & Units

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All physical quantities in the codebase use the same unit system: **ms for time, mV for voltage, nF for capacitance, μ S for conductance, nA for current, Hz for rates**. Time fields carry an explicit *_ms* suffix (*sim_ms*, *ref_ms_E*, *tau_gaba*); CLI flags follow the same convention (*-t-ms 600*). A Δt of 1 means 1 ms, not 1 s.

Quantities

Quantity	Unit	Typical value	Variable
Integration step	ms	0.25	<i>dt</i> , <i>DT_MS</i>
Simulation length	ms	600	<i>sim_ms</i>
Membrane time constant	ms	20 (E), 5 (I)	<i>tau_m_E</i> , <i>tau_m_I</i>
Refractory period	ms	3 (E), 1.5 (I)	<i>ref_ms_E</i> , <i>ref_ms_I</i>
AMPA decay	ms	2	<i>tau_ampa</i>
GABA decay	ms	9	<i>tau_gaba</i>
Resting / leak potential	mV	−65	<i>E_L</i>
Spike threshold	mV	−50	<i>V_th</i>
Reset potential	mV	−65	<i>V_reset</i>
AMPA reversal	mV	0	<i>E_e</i>
GABA reversal	mV	−80	<i>E_i</i>
Membrane capacitance	nF	1.0 (E), 0.5 (I)	<i>C_m_E</i> , <i>C_m_I</i>
Leak conductance	μ S	0.05 (E), 0.1 (I)	<i>g_L_E</i> , <i>g_L_I</i>
External drive	μ S	0.0006 (async), 0.003 (PING)	<i>t_e_async</i> , <i>t_e_ping</i>
Input current (CUBA)	nA	20 per spike	<i>input_scale</i>
CUBA weight std	nA	32	<i>W_STD_CUBA</i>
Max input rate	Hz	25	<i>max_rate_hz</i>
Population firing rate	Hz	20–80	<i>r_E</i> , <i>r_I</i>
Gamma frequency	Hz	30–80	<i>f_0</i>

COBA / PING biophysical constants

Used by COBA and PING. Values follow neuroscience conventions (cf. Dayan & Abbott, Gerstner *Neuronal Dynamics*); the E:I asymmetry in $\tau_m, C_m, g_L, \tau_{\text{ref}}$ produces the timescale separation that makes PING dynamics possible.

Parameter	E population	I population
τ_m (ms)	20	5
C_m (nF)	1.0	0.5
g_L (μ S)	0.05	0.1
τ_{ref} (ms)	3	1.5
E_L (mV)	−65	−65
V_{th} (mV)	−50	−50
V_{reset} (mV)	−65	−65

E_e (mV, reversal)	0	0
E_i (mV, reversal)	-80	-80

Synapse time constants: $\tau_{\text{AMPA}} = 2$ ms (excitation), $\tau_{\text{GABA}} = 9$ ms (inhibition). These set the ceiling on PING's Δt -stability: once $\Delta t \gtrsim \tau_{\text{GABA}}$ the $E \rightarrow I \rightarrow E$ loop cannot complete within one step.

Internal consistency

The chosen units are self-consistent — no conversion factors appear in the integration code. Two equations carry the whole system.

The membrane time constant is $\tau = C/g$. With C in nF and g in μS ,

$$\tau_{[\text{ms}]} = \frac{C_{[\text{nF}]}}{g_{[\mu\text{S}]}}$$

so $C_m = 1$ nF and $g_L = 0.05$ μS give $\tau_m = 20$ ms directly.

The LIF voltage update is $dv = (\Delta t/C)(-g_L(v - E_L) + I)$. With Δt in ms, C in nF, v, E in mV, g in μS , and I in nA,

$$dv_{[\text{mV}]} = \frac{\Delta t_{[\text{ms}]}}{C_{[\text{nF}]}} \cdot I_{[\text{nA}]}$$

because $\text{ms} \cdot \text{nA} / \text{nF} = \text{mV}$ exactly.

Conductance-current products share the same ledger: $g(v - E)$ is $\mu\text{S} \times \text{mV} = \text{nA}$, so synaptic currents fold into I alongside any direct input current without a scale factor.

Why not SI?

Pure SI (F, S, V, A, s) forces every value to a large negative exponent — $C_m = 10^{-9}$ F, $g_L = 5 \times 10^{-8}$ S, $\Delta t = 2.5 \times 10^{-4}$ s. The neuroscience convention (ms, mV, nF, μS , nA) keeps every typical value between 10^{-3} and 10^2 , which makes numerical debugging and human intuition faster. The tradeoff is that readers have to trust the unit consistency rather than verify it by plugging into SI formulas — this page is here so that trust is auditable.